

- (b) One of the compounds found in petrol is octane, C_8H_{18} .

$$A_r(\text{C}) = 12$$

- (i) Give the **total** relative mass of the carbon atoms in one molecule of octane. [1]

Mass =

- (ii) The relative molecular mass of octane is **114**. Use your answer to part (i) to calculate the percentage by mass of carbon in octane. [1]

Percentage = %



(c) Fuel, heat and oxygen are the three factors needed to start and maintain a fire.

The photographs show three different methods used to put out fires or stop them from spreading.

Method	Reason
	removes heat
	removes oxygen
	removes fuel

(i) Draw **one** line from each method to the reason it puts out a fire. [2]

(ii) Underline the type of fire that would be unsafe to put out using water. [1]

bonfire electrical fire forest fire

